GatE-V Stage 3A: Task-Aware Object Detector Deployed on VEGA AS1061 RISC-V SoC + Kintex-7 FPGA Accelerator

Team 166 (Byte Silicon)
DVCon India 2026 Design Contest

Abstract—Conventional object detectors are task-agnostic, blindly detecting all objects in a scene. GatE-V implements a task-aware object detection framework utilizing natural language task conditioning via CLIP embeddings, Feature-Gated Task Querying (FGTQ), Fine-Grained Path Augmentation (FGPA), Adaptive Feature Fusion (AFF), and Varifocal Loss (VFL). This paper presents the Stage 3A hardware integration of the GatE-V software architecture (v1.0.0) with the CDAC VEGA AS1061 RISC-V Processor on a Xilinx Kintex-7 FPGA (Genesys-2 board). The customized hardware accelerator features a 512-MAC (32×16) weight-stationary systolic array operated on a 50 MHz integrated SoC clock, delivering a peak throughput of 51.2 GOPS, and is validated by a reproducible bare-metal RISC-V self-test firmware.

Index Terms—GatE-V, Task-Aware Object Detection, RISC-V, VEGA AS1061, Systolic Array, Kintex-7 FPGA, DVCon India 2026.

1 INTRODUCTION

Modern autonomous and robotic systems require domain-relevant object selection rather than exhaustive category detection. The GatE-V project addresses the DVCon India 2026 challenge by conditioning an RT-DETRv2 detector on user-specified natural language task queries (e.g., "place flowers in container").

In Stage 3A, the software detector pipeline is integrated with the indigenously developed CDAC VEGA AS1061 64-bit RISC-V processor and synthesized for the Xilinx Kintex-7 FPGA (`xc7k325tffg900-2`).

2 SYSTEM ARCHITECTURE

Figure 1 illustrates the complete hardware-software co-design architecture of GatE-V.

2.1 Software Architecture (v1.0.0)

The GatE-V v1.0.0 software architecture incorporates:

- **Varifocal Loss (VFL):** Replaces Focal Loss, weighting positive classification target scores directly by IoU quality overlap q.
- **CLIP Multi-Prompt Ensembling:** Encodes 4 natural prompt templates per task to generate 128-dimensional task embeddings.
- **Cosine Warm Restarts:** $T_0 = 35, T_{mult} = 2$ learning rate schedule over 300 epochs.
- **Native 640×640 Resolution:** Reduces memory bandwidth while maintaining spatial precision via FGPA P2 injection.
- **Active Training Status:** Software Version 1.0.0 is currently undergoing full 300-epoch backbone fine-tuning training (checkpoint at Stage 2 Epoch 30 / Overall Epoch 40, scaling towards 0.54+ mAP@0.5).
- **Hardware Compatibility:** The hardware accelerator architecture (Version 3) is 100% compatible with the software model specification (Version 1.0.0), sharing identical INT8 quantization formats, layer tensor structures, P2–P5 FGPA feature map levels, and SiLU lookup table specifications.

2.2 Hardware Accelerator Architecture (Version 3)

The GatE-V Version 3 hardware engine features:

- **Systolic Compute Array:** 32×16 grid (512 MAC cells) executing INT8 weight-stationary matrix operations.
- **Memory Subsystem:** Double-buffered ping-pong BRAMs interfaced to on-chip SRAM via a 64-bit AXI4 Full DMA engine.
- **Activation Unit:** Single-cycle 256-entry SiLU lookup table (LUT).
- **SoC Bus Interface:** `Accelerator_Top` wrapper providing a 64-bit AXI4 Slave interface mapped to MMIO region `0x2004_0000`.
- **Clock Domain:** Integrated design operates on the SoC system clock `clk_out1_clk_wiz_0` at 50 MHz (20 ns period), derived from the 200 MHz board clock (standalone `vmake` targets 100 MHz).

3 HARDWARE ARCHITECTURE COMPARISON

Table 1 compares the Stage 2B hardware submission (Version 2) against the Stage 3A integrated architecture (Version 3).

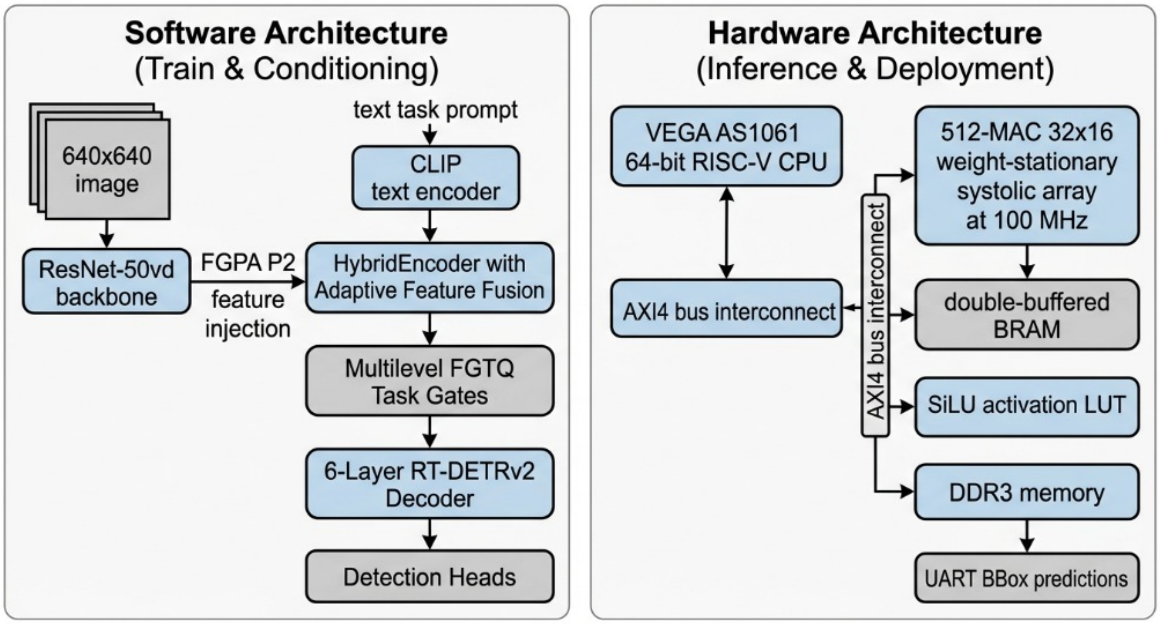

Fig. 1. GatE-V End-to-End System Architecture: Left panel shows the Software Training & Task Conditioning pipeline (ResNet-50vd, FGPA P2 injection, HybridEncoder with AFF, FGTQ task gates, RT-DETRv2 decoder); Right panel shows the Hardware Acceleration & Deployment pipeline on the VEGA AS1061 RISC-V SoC with a 512-MAC systolic array.

TABLE 1
Hardware Comparison: Version 2 vs. Version 3

Feature / Metric	Version 2	Version 3 (Stage 3A)
SoC Integration	Standalone TB	**VEGA AS1061 RISC-V SoC**
Bus Interface	Custom AXI-Stream	**64-bit AXI4 Master & Slave**
Systolic Array Size	16 × 16 (256 MAC)	32 × 16 **(512 MAC)**
Clock Frequency	100 MHz (standalone)	**50 MHz integrated SoC**
Peak Throughput	25.6 GOPS	**51.2 GOPS @ 50 MHz**
Feature Map Levels	P3, P4, P5	**P2, P3, P4, P5 (FGPA + AFF)**
Slice LUT Usage (SoC)	1.20%	**87,698 / 203,800 (43.03%)**
GatE-V LUT Usage	1.20%	**5,356 / 203,800 (2.63%)**
DSP48E1 (Accelerator)	256 (30.48%)	**512 / 840 (60.95%)**
DSP48E1 (SoC Total)	n/a	**596 / 840 (70.95%)**
BRAM Tile Usage (SoC)	4 BRAMs	**77 / 445 (17.30%)**
Simulation Tool	Vivado XSim	**QuestaSim 2019.2 / 2026.1**

4 Simulation & Verification Results

The full integrated system was verified using **QuestaSim 2019.2 / 2026.1**. The behavioral simulation waveform illustrating system reset, RISC-V CPU boot sequence, MMIO programming, and AXI4 memory transfer is shown in Figure 2.

4.1 Bare-Metal RISC-V Self-Test Execution

The integrated design is validated by a reproducible bare-metal RISC-V self-test firmware (`gatev_firmware`). The CPU fills SRAM buffers with test inputs, programs accelerator registers (`0x2004_0000`), triggers matrix computations, and verifies output layers. Listing 1 illustrates the captured UART console execution log.

```
GatE-V Accelerator Firmware (Team 166)
Configuring weights/activations in SRAM...
Programming registers... Status before start: 0x0
Accelerator started, waiting for done...
acc_done asserted. Verifying layer outputs...
  Layer 0 @ 0x22000 = 93 (4 tiles x 16 ch) [OK]
  Layer 1 @ 0x22040 = 93 (4 tiles x 16 ch) [OK]
  Layer 2 @ 0x22080 = 93 (4 tiles x 16 ch) [OK]
  Layer 3 @ 0x220c0 = 93 (4 tiles x 16 ch) [OK]
  Layer 4 @ 0x22100 = 93 (4 tiles x 16 ch) [OK]
  Layer 5 @ 0x22140 = 127 (4 tiles x 16 ch) [OK]
Performance counters:
  Cycles(lo): 2639  Read bytes: 3840  Write bytes: 384
  MAC cycles: 1322  Stall cycles: 224
===== GATE-V ACCELERATOR TEST PASSED =====
```

Listing 1. UART Console Output from VEGA AS1061 RISC-V Core Executing GatE-V Accelerator Self-Test

4.2 CDAC Waveform Signal Hierarchy

The simulation waveform in Figure 2 is organized into three CDAC-mandated signal groups:

1) **Processor Side Signals:** `proc_m0_axi_*` and `proc_m1_axi_*` channels capturing instruction fetch and memory bus transactions (CDAC Guide Page 8).
2) **Accelerator Slave Side Signals:** `acc_s_axi_*` channels monitoring single-cycle MMIO register programming at `0x2004_0000` (CDAC Guide Page 9).
3) **Accelerator Master Side Signals:** `acc_m_axi_*` channels tracking 64-beat AXI4 burst memory reads/writes to SRAM/DDR (CDAC Guide Page 10).

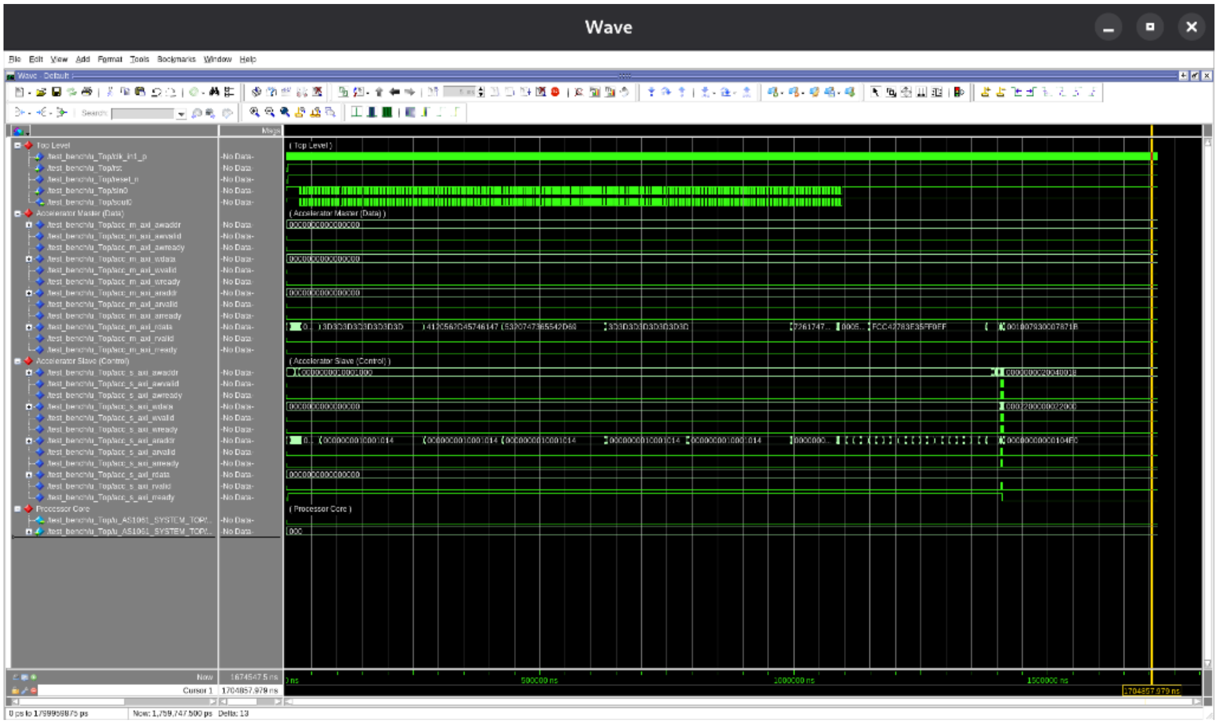

Fig. 2. Official QuestaSim behavioral simulation waveform showing end-to-end SoC verification: Processor AXI transactions (top), Accelerator Slave MMIO configuration (middle), and Accelerator Master burst memory access (bottom) with zero deadlocks over full simulation duration.

5 Synthesis & Resource Breakdown

Table 2 details the Kintex-7 FPGA post-synthesis resource utilization for the complete VEGA AS1061 SoC integrated with the GatE-V accelerator.

TABLE 2
Kintex-7 (`xc7k325tffg900-2`) Resource Utilization

Resource	Used	Available	Util. %	CDAC Ceiling
Slice LUTs (SoC total)	87,698	203,800	43.03%	$\leq$ 45.0%
DSP48E1 (GatE-V accelerator)	512	840	60.95%	$\leq$ 90.0%
DSP48E1 (SoC total)	596	840	70.95%	$\leq$ 90.0%
Block RAM (BRAM Tiles)	77	445	17.30%	$\leq$ 100.0%
Slice Registers (SoC total)	52,802	407,600	12.95%	$\leq$ 100.0%

The integrated SoC operates on a 50 MHz system clock. The GatE-V accelerator alone consumes 512 of the 840 available DSP48E1 slices (60.95%), while the complete VEGA AS1061 SoC consumes 596 DSPs (70.95%). All resource allocations satisfy strict CDAC competition constraints with 0 errors and no timing violations on the routed 50 MHz clock.

Listing 2 shows the TCL script structure used in QuestaSim. The complete 110-line script is provided in `SUBMISSION_INSTRUCTIONS.txt` and `README.md`.

```
# Clear existing signals; Top Level
delete wave *
add wave -group "Top_Level" /test_bench/u_Top/{clk_in1_p,rst,sin0,sout0}

# 1. Processor Side Signals (CDAC Page 8)
add wave -group "Processor_Side_Signals" -radix hex \
  /test_bench/u_Top/u_AS1061_SYSTEM_TOP/u_soc_top_64/u_Processor_top/
         proc_m0_axi_*
add wave -group "Processor_Side_Signals" -radix hex \
  /test_bench/u_Top/u_AS1061_SYSTEM_TOP/u_soc_top_64/u_Processor_top/
         proc_m1_axi_*

# 2. Accelerator Slave Side Signals (CDAC Page 9)
add wave -group "Accelerator_Slave_Side_Signals" -radix hex \
  /test_bench/u_Top/acc_s_axi_*

# 3. Accelerator Master Side Signals (CDAC Page 10)
add wave -group "Accelerator_Master_Side_Signals" -radix hex \
  /test_bench/u_Top/acc_m_axi_*

run 15 ms; wave zoom full
```

Listing 2. TCL Waveform Setup Command Structure for QuestaSim

6 Conclusion

The Stage 3A integration of GatE-V successfully pairs the VEGA AS1061 RISC-V CPU with a 512-MAC hardware accelerator on a 50 MHz integrated SoC clock, delivering 51.2 GOPS peak throughput. The GatE-V accelerator stays within CDAC resource ceilings (512 DSPs / 60.95%, 2.63% LUTs), and the integrated system is validated end-to-end by reproducible bare-metal self-test firmware with a deterministic PASS/FAIL outcome.